Kenji Watanabe

Machine Learning Researcher

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Research Summary

PhD candidate researching efficient transformer architectures for low-resource speech recognition. Author of 3 peer-reviewed papers. Seeking research engineer roles in applied ML.

Education

PhD, Computer Science Sep 2021 – Present
University of Example Toronto

- Thesis: Parameter-efficient adaptation of speech transformers.

M.Eng., Information Science Apr 2019 – Mar 2021
Sample Institute of Science Osaka, Japan

Research Experience

Research Intern May 2024 – Aug 2024
Litware AI Lab · Internship Montreal, QC

- Implemented low-rank adapters for a 300M-parameter speech model, matching full fine-tuning accuracy with 2% of trainable parameters.
- Open-sourced the training code and evaluation scripts.

Technologies: PyTorch, JAX

Graduate Research Assistant Sep 2021 – Present
University of Example Toronto · Part-time

- Teaching assistant for the graduate Machine Learning course (2 terms).

Publications

Adapter Tuning for Low-Resource Speech Recognition 2024
Proceedings of the Example Speech Conference
K. Watanabe, A. Example

Pruning Attention Heads in Streaming ASR 2023
Example Workshop on Efficient ML
K. Watanabe, B. Sample, A. Example

Awards

Graduate Research Fellowship 2021
University of Example Toronto

Skills

ML: PyTorch, JAX, Hugging Face Transformers, Speech recognition
Programming: Python, C++, CUDA